CW-ERM: Improving Autonomous Driving Planning with Closed-loop Weighted Empirical Risk Minimization [1]

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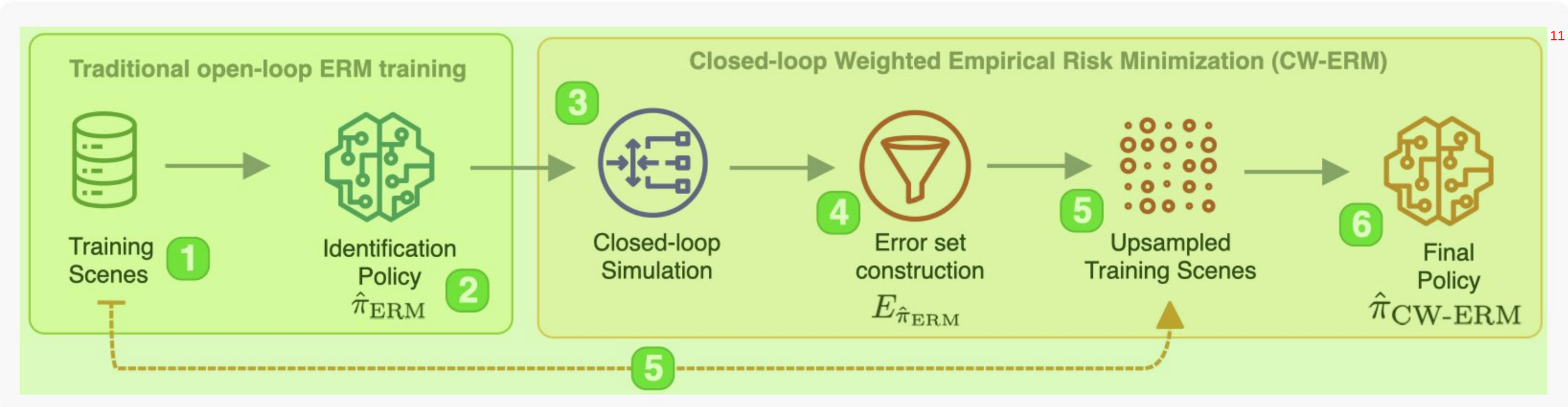
NEURAL INFORMATION PROCESSING SYSTEMS [5]

Main Ideas and Motivation [6]

- Behavior Cloning (BC) algorithms for training self-driving vehicles via pure Empirical Risk Minimization (ERM) biases policy towards matching open-loop behavior [7]
- We debias the policy network by identifying important samples via a closed-loop evaluator
- Training in three phases [8]
  - Open-loop identification phase
  - Error set construction and evaluation phase
  - Upsampling training phase

Our Contributions [9]

- Motivate and propose a technique that leverages closed-loop evaluation metrics acquired from policy rollouts in a simulator to debias the policy network and reduce the distributional differences between training (open-loop) and inference time (closed-loop); [10]
- Evaluate CW-ERM on a challenging urban driving dataset in a closed-loop fashion to show that our method, although simple to implement, yields significant improvements in closed-loop performance without requiring complex and computationally expensive closed-loop training methods;
- Show an important connection of our method to a family of methods that addresses covariate shift through density ratio estimation.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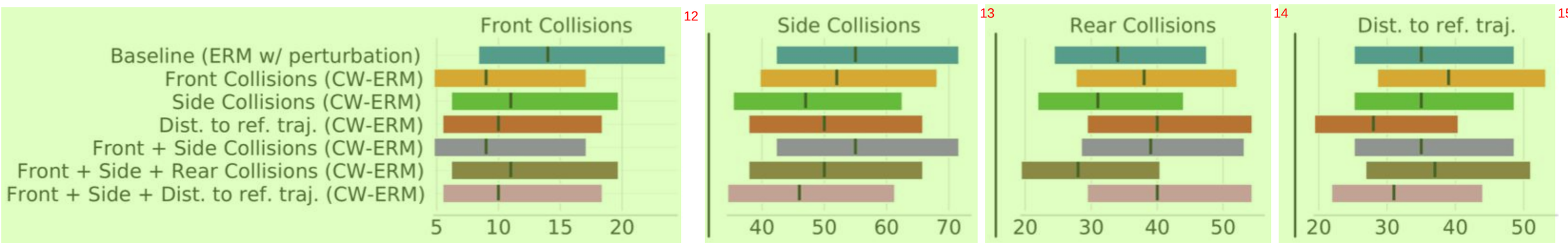
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Methodology [17]

Stage 1 (Identification Policy) [18]

- Train policy network via BC in open-loop using ERM, get $\hat{\pi}_{ERM}$ [19]

Stage 2 (Closed-Loop Simulation) [20]

- Perform rollouts using $\hat{\pi}_{ERM}$ in closed-loop simulator [21]
- Collect closed-loop metrics and identify the error set

$$E_{\hat{\pi}_{ERM}} = \{(s_i \,.\, a_i) \text{ s.t. } C(s_i, a_i) > 0\},$$ [22]

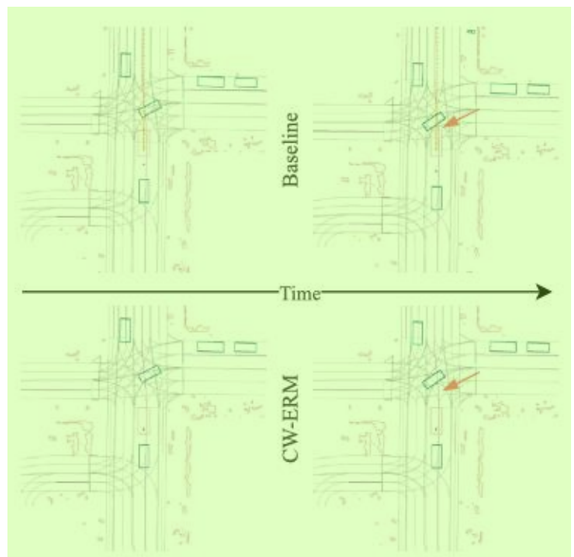
where $C(s_i, a_i)$ is a cost (e.g. collisions) incurred during the closed-loop rollouts. [23]

Stage 3 (Final Policy) [24]

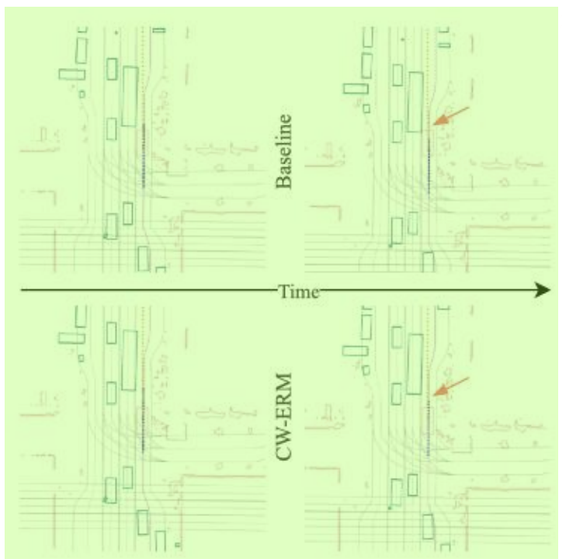
- Train new policy where scenes belonging to $E_{\hat{\pi}_{ERM}}$ are unweighted by a factor $w(\cdot)$. [25]

$$\arg\min_{\pi \in \Pi} \mathbb{E}_{s \sim d_{\pi^*}, a \sim \pi^*(s)} [w(E_{\hat{\pi}_{ERM}}, s) \ell(s, a, \pi)]$$ [26]

Collision Avoidance Examples [16]


 [27]

Front collisions [28]


 [29]

Side collisions [30]

References [31]

1. Liu et al. (2021). Just Train Twice: Improving Group Robustness without Training Group Information. *International Conference on Machine Learning*. [32]
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[33]

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